

Retail Sales Data Agent Project

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Introduction & Objectives

This project is about building a Data Agent on databricks for a retail sales dataset, where the Data Agent will demonstrate performance and customer demographics analysis.

The core objective of this project is to analyze transactional data from a retail store over a one-year period (from January 1, 2023, to January 1, 2024) to uncover actionable insights regarding operational performance and customer purchasing behavior.

The analysis includes:

- **Sales Performance & Trend Analysis:** Tracking how revenue behaves over time to identify seasonal spikes, monthly growth rates, and top-performing periods. For example, calculating total revenue, which sits at 456,000 across 1,000 transactions.
- **Product Category Insights:** Evaluating sales across major product categories (Beauty, Clothing, and Electronics) to determine which categories drive the highest volume and revenue.
- **Customer Segmentation & Demographics:** Linking sales behavior to customer profiles. By exploring characteristics like Age (ranging from 18 to 64 years) and Gender, we can discover which demographics are the most valuable to the business and what products they prefer.

Dataset Description

The dataset contains transactional records capturing retail purchase activity over a one-year period. It consists of 1,000 unique transaction records across 9 distinct columns.

Here is what each specific column represents and the type of data it holds:

Column Name	Data Type	Description & Business Meaning	Examples from Data
Transaction ID	Integer	A unique identifier assigned to every individual transaction. Used as a primary key to track distinct purchases.	1, 2, 3
Date	Object	The exact date the purchase took place. Essential for tracking monthly sales trends, seasonality, and time-series forecasting.	2023-11-24
Customer ID	Object	A unique code assigned to each customer. This allows analysts to track repeat purchase behavior and map out customer journeys.	CUST001
Gender	Object	The customer's gender (Male or Female). Used for basic demographic profiling and targeted marketing.	Female
Age	Integer	The age of the customer at the time of purchase (ranging from 18 to 64 years old). Crucial for age-based market segmentation.	34

Product Category	Object	The broad classification of the item sold. The dataset focuses on three main retail segments: Beauty, Clothing, and Electronics.	Beauty
Quantity	Integer	The number of units of the product purchased in that specific transaction.	3
Price Per Unit	Integer	The cost of a single unit of the product.	50
Total Amount	Integer	The total revenue generated from the transaction.	150

Screenshots:

Uploaded retail sales dataset

The screenshot displays the Databricks workspace interface. On the left, a sidebar contains navigation links: Home, Learn, Workspace, Recents, Catalog (selected), Jobs & Pipelines, Compute, Discover (Beta), Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, and Query History. The main area is titled 'Catalog' and shows a tree view of the data catalog. The 'retail_sales_dataset' table is selected under the path: My organization > workspace > system > database > retail > default > information_schema > sales > Tables (1). The table's description is 'Created by the file upload UI'. A table of columns is displayed with the following details:

Column	Type	Comment	Tags	Column masking
Transaction ID	bigint			
Date	date			
Customer ID	string			
Gender	string			
Age	bigint			
Product Category	string			
Quantity	bigint			
Price per Unit	bigint			
Total Amount	bigint			

This is the Agent about screenshot page

AboutDataInstructions

About this space

Name

Retail Business Intelligence Analyst

Owner

pansybanetsi@gmail.com

Space ID

01f157776d8f1fb29ae9968a549b7abb

Warehouse

 Serverless Starter Warehouse

Tags

Thumbnail



Description

Retail Sales Analysis Agent

This space provides insights into retail sales transactions, enabling analysis of customer behavior, product performance, and revenue trends.

This is the Agent instructions screenshot

AboutDataInstructions

TextJoinsSQL Express...SQL Queries

General Instructions

Add general instructions on how you want this Genie space to behave.

Retail Business Intelligence Analyst

Role: As a Retail Business Intelligence Analyst, your primary key role is to evaluate overall business performance using the retail sales dataset , empowering the CEO to make strategic and growth-focused decisions.

Objective: The primary objective of this retail dataset is to analyze transactional records to uncover valuable insights into customer purchasing behavior, sales performance, and product trends. By exploring how customer demographics like age and gender correlate with metrics like transaction amounts and quantities, businesses can make data-driven decisions to optimize inventory, improve targeted marketing, and maximize overall profitability.

Core Task: Demographic Segmentation & Behavior Analysis: Analyzing how different customer groups (segmented by gender and age cohorts) spend across product categories like Beauty, Clothing, and Electronics to understand consumer preferences.

Sales Trend & Time-Series Analysis: Tracking the transaction dates to identify historical sales patterns, seasonal fluctuations, peak shopping periods, and overall revenue growth over time.

Product Performance & Revenue Metrics: Evaluating product categories based on quantities sold, pricing structures, and total contribution to revenue to assist with pricing strategies and inventory forecasting.

Ranking Logic: Rank by net revenue descending. Break ties using unit volume, then alphabetical order. Never invent ranking signals not present in the source tables.

Output Rules: Structure every response as: Headline (one sentence) · Key Numbers table (≤ 10 rows) · Context paragraph (< 60 words). Format currency as ZAR, rounded to thousands.

Audience & Tone: Your audience is sales leaders and finance partners – numerate but not technical. Use a confident, concise business tone. Avoid SQL terminology and engineering jargon.

Constraints: Use only fact_sales, fact_refunds, dim_product, dim_customer. Never query archived sales or raw /staging /tmp

tables. Exclude internal test accounts. Never provide forecasts.

Error & Ambiguity Handling: If data is missing, state the limitation and offer best-effort. If a question is ambiguous, state your assumption. If a query returns zero rows, say 'no matching records' – never report it as zero.

The 10 Questions that I asked the Agent:

Are there specific months where sales experience specific spikes or drops and which product spike the most and least?

> Thinking complete

May Shows the Highest Spike, September the Steepest Drop

May 2023 experienced the highest revenue spike at \$53,150, while September 2023 saw the steepest drop at \$23,620—representing a 125% increase above the lowest month. ¹⁶

Monthly Sales Patterns

The year shows significant volatility with three notable peaks: May (\$53,150), October (\$46,580), and December (\$44,690). The lowest performing months were September (\$23,620), March (\$28,990), and April (\$33,870). ¹⁶

Are customers buying multiple units quantity of certain product categories more frequently than others?

> Thinking complete

Beauty Products Lead in Multi-Unit Purchases

Yes, customers buy multiple units more frequently for Beauty products (75.9% of transactions) compared to Clothing (74.9%) and Electronics (73.4%), though the differences are modest. [15](#)

Key Findings

Across all three categories, **multi-unit purchases dominate**—approximately 3 out of 4 transactions involve buying 2 or more units. Beauty leads with 233 multi-unit transactions out of 307 total, while Electronics has the lowest multi-unit rate at 73.4% (251 out of 342 transactions). [15](#)

The average quantity per transaction is remarkably consistent: **Clothing averages 2.55 units, Beauty 2.51 units, and Electronics 2.48 units.** All categories show purchases ranging from 1 to 4 units per transaction. [15](#)

Which demographic segments in terms of age group, demonstrate the least purchases

Thinking complete

Young Adults (18-24) Make the Fewest Purchases

The 18-24 age group demonstrates the least purchases with only 149 transactions, representing 14.9% of all transactions—significantly lower than any other age segment. [12](#)

Key Findings

The 45-54 age group leads with 225 transactions, followed by 55-64 (216 transactions), 35-44 (207 transactions), and 25-34 (203 transactions). Young Adults trail by a substantial margin—**51% fewer transactions than the highest-performing segment.** [12](#)

Despite having the lowest transaction volume, Young Adults show the **highest average transaction value at \$501**, compared to \$417-\$478 for other age groups. This suggests they make fewer but higher-value purchases. [12](#)

According to the dataset, on which day do customers make more transactions and what products are being purchased?

> Thinking complete

Tuesday Has the Most Transactions with Balanced Product Mix

Tuesday leads with 161 transactions (16.1% of all transactions), followed by Saturday (150) and Monday (146). Thursday has the fewest at 123 transactions. [7](#)

Products Purchased on Tuesday

On Tuesday, customers purchase a **well-balanced mix across all three product categories:** Beauty (56 transactions, 34.8%), Clothing (55 transactions, 34.2%), and Electronics (50 transactions, 31.1%). [8](#)

Which product category contributes the highest overall percentage of sales?

Thinking complete

Electronics Leads with 34.4% of Sales

Electronics contributes the highest percentage at 34.4% of total revenue (\$156,905), narrowly edging out Clothing at 34.1% (\$155,580) and Beauty at 31.5% (\$143,515). [4](#)

The three product categories are remarkably balanced, with only a 2.9 percentage point spread between the highest (Electronics) and lowest (Beauty) performers. All three categories generate between \$143K-\$157K in revenue from 307-351 transactions each. [4](#)

What price items drive the bulk of our revenue?

> Thinking complete

High-Price Items Drive Revenue

Premium-priced items at \$500 and \$300 generate 88% of total revenue, despite representing only 40% of transactions. [3](#)

Key Findings

The \$500 price point alone accounts for 54.3% of all revenue (\$247,500 out of \$456,000 total), making it your single most important revenue driver. [3](#)

Adding the \$300 price point brings cumulative revenue to 88.4%, with these two premium tiers generating \$402,900 combined from just 396 transactions. [3](#)

According to this dataset is this enterprise performing very well?

Thinking complete

No — This Enterprise Has Critical Performance Issues

The business is not performing well despite generating \$456,000 in revenue. While some surface metrics appear reasonable, the enterprise faces severe structural problems that threaten long-term viability, most critically a 0% customer retention rate. [29,30](#)

Are there any product implications that the business needs to focus on to prevent the sales?

> Thinking complete

Yes — Critical Product Issues Are Preventing Sales

The business faces five major product-related problems that are directly preventing sales and threatening viability. These issues span pricing strategy, product mix, seasonal management, and customer targeting.

1. Catastrophic Electronics Volatility — Immediate Crisis

Electronics revenue collapsed 85% in March (from \$15,465 in February to just \$3,380), representing a \$19,865 swing between best and worst months—2x more volatile than other categories. This extreme instability suggests severe supply chain issues, inventory mismanagement, or product availability problems. [38,44](#)

Based on the high-volume sales periods identified in the date column, when should the operations team optimize staffing and inventory levels?

> Thinking complete

Optimize for May, October, and Early Week (Tuesday-Monday)

The operations team should prioritize staffing and inventory for May (peak month), October (second peak), and Tuesday-Monday (highest transaction days). May sees 105 transactions and 259 units sold—62% higher volume than the slowest month (September). [26](#)

Monthly Optimization Strategy

High-Volume Months (Increase Staffing & Inventory 20-30%)

Based on the dataset, what percentage of our customer base are repeat buyers versus one-time shoppers?

> Thinking complete

100% One-Time Shoppers — No Repeat Buyers

The dataset shows 0% repeat buyers and 100% one-time shoppers. All 1,000 unique customers made exactly one purchase each, with no customer returning for a second transaction. [22,23](#)

Key Findings

The data contains 1,000 unique customers and 1,000 transactions, resulting in an average of exactly 1.0 transactions per customer. Every customer ID appears only once in the dataset. [23](#)

Each customer's first purchase date equals their last purchase date, confirming no repeat purchases occurred during the data period (2023). [21](#)

Validation of 3 answers:

1. Which categories do customers buy the most?
2. Who are the typical customers?
3. Which days/months usually have the most transactions?

Databricks syntax:

The screenshot shows the Databricks SQL Editor interface. On the left is a sidebar with navigation options: Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover (Beta), and Marketplace. Below these are sections for SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, and Query History. The main area displays a SQL query:

```
1 select * from `retail`.`sales`.`retail_sales_dataset` limit 100;
2 SELECT 'Product Category', SUM('Quantity') AS Total_Quantity
3 FROM retail.sales.retail_sales_dataset
4 GROUP BY 'Product Category'
5 ORDER BY Total_Quantity DESC;
```

Below the query, there is a table with 2 columns: Product Category and Total_Quantity. The table shows 1 row of data:

Product Category	Total_Quantity
Product Category	2514

At the bottom, it indicates "1 row | 10.56s runtime" and "See performance (2)". The interface also shows "Run all (1000)" and "16 hours ago (11s)" status.

Business Recommendations:

Based on the retail sales data, different age groups show preferences for what they buy, paving the way for hyper-targeted product pairings. Analysis shows that under 25s spend the most heavily on Beauty (\$31,280), under 25-35 years old are major drivers of Clothing (\$39,975) and over 40 years old lean heavily toward Electronics. Creating cross-category cross-selling campaigns based on login or loyalty card demographics for users under 35, bundle Clothing & Beauty items together at a slight discount during checkout, for users over 45, pitch practical Electronics accessories (like premium chargers or warranties) to maximize their cart totals.

Conclusion:

While executing this project under tight timelines was a significant challenge for a transitioning tech professional, it provided a steep and rewarding learning curve. Securing a one-week project window in the future will allow for the deep-dive research required to maximize development quality and insights.